BGIN Block#11 Meeting Report

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Tokenization Initiatives

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BGIN Block#11

Tokenization Initiatives

Date: Monday, October 21, 2024

Location: Washington DC, United States

Duration: Approx. 90 minutes

Summary

Session Overview:

- Focus on organizational tokenization initiatives led by central banks, regulatory authorities, and financial institutions.
- Presentations from JPMorgan (JPM) and the Bank of Japan (BOJ) on interoperability challenges in tokenization projects.
- Introduction to various interoperability frameworks, including connectivity methods.

JPMorgan's Presentation on Connectivity & Interoperability:

- **Importance of Interoperability**: Key to connecting assets and payments within economic transactions, such as delivery versus payment (DVP).
- **Atomic Swaps**: Explanation of atomic swaps, where transactions on the same blockchain platform confirm simultaneously; contrast with complex multi-platform DVP.
- **Two-Phase Commit Protocol**: Used for cross-platform DVP, involving cash and asset locks to enable synchronized settlement.
- **Connectivity Models**:
    - API-level connectivity on the application layer.
    - Blockchain-to-blockchain connectivity, emphasizing bridges to reduce inefficiencies.
    - Smart contract composability for direct interaction within shared ledgers.
- **Examples & Projects**:
    - Live transaction with Siemens using payment synchronization for faster settlement.
    - Global Ledger 1 (GL1): Collaborative platform for shared ledger integration among large banks.

GL1 Platform:

- **Concept**: A unified platform with various banks (e.g., JPMorgan, MUFG, Citibank) aimed at efficiency and reduced costs through shared infrastructure.

- **Challenges**:
  - Governance restrictions prevent banks from using permissionless networks.
  - Preference for public-permissioned models to retain KYC and regulatory controls.
- **Interoperability**: Multiple methods to connect apps and blockchains to achieve efficient cross-platform interactions.
- **Future Direction**: Potential for convergence between permissioned and permissionless networks over time.

Questions from Participants:

- **Privacy and Security**:
  - Privacy issues in multi-party smart contract interactions; difficulty in restricting information to relevant parties.
  - Technical challenges of orchestrating multiple smart contracts.
- **Technical Standards**:
  - Need for standards for tokenized assets, especially with varied pricing and liquidity requirements.
- **Regulatory Access and Data Sharing**:
  - Discussion on data sharing with regulators and potential challenges due to moral hazard.
- **Cybersecurity**:
  - High-level cryptographic review within projects; specific security details handled by specialized teams.

Bank of Japan's involvement in Project Agora:

- **Objective**: Collaborate with BIS and other central banks to integrate tokenized commercial and central bank money on a shared programmable ledger.
- **Approach**:
  - Public-private collaboration to improve cross-border payments with a focus on AML/CFT compliance and transaction transparency.
  - Focus on feasibility studies and prototyping multi-currency ledger systems by 2025.
- **ASAP Model**: Framework (Assets, Services, Applications, Platform) guiding both GL1 and Project Agora's design, differentiating their focuses.

IETF Standards on Secure Asset Transfer:

- **Role of IETF**: Develops interoperability standards for common functions in DLTs; work focuses on universal asset schema and asset transfer protocols.
- **Participation & Goals**: Encourages cross-sector collaboration to standardize asset recognition and transfer across networks.

Final Remarks:

- Emphasis on the importance of technical interoperability standards and collaborative frameworks to support innovation while ensuring regulatory compliance and secure cross-border transactions.
- Encouragement for more organizations to participate in standard-setting bodies, such as IETF, to streamline cross-platform interoperability.

Transcript

All right. So the first session is the organization initiatives. And yeah, there are so many tokenization initiatives led by central banks, regulatory authorities, and leading financial institutions these days. And so we're going to focus, we're going to have good presentations about each organization initiative by JPM and also like the BOJ, the Bank of Japan. And we're going to focus on the interoperability challenges all of the tokenization initiatives currently have. And before talking about interoperability challenges, a little bit technical, but before talking about this, we would like people here to make a presentation about each project first. And so can you kick off talking about initiatives you are working on?

Thank you so much. Maybe just a quick introduction. I'm the Global Head of Business Architecture for Onyx Coin Systems. Essentially what we do is very much on blockchain payments, digital currencies. So today the topic is talking about connectivity and interoperability. So before we share some of the work that we are doing, I just wanted to share a framework of how we are thinking about connectivity and interoperability. I hope that helps also to frame the discussion for later.

So maybe the first step, why are we so interested in connectivity interoperability? From our perspective, payments is always one side of an economic transaction. You always often have another side to it. So the curious example is DVP of securities and payments, because essentially you have a transfer of assets on one side, you have a corresponding payment on the other side. And you always want some ability to tie these two transactions together.

Now, generally, when we think about connectivity, there's really two ways of doing it. If they are on a common platform itself, we are able to do what we call an atomic swap. But if they are not on the common platform, we can still achieve DVP, but through a much more complicated means. So when we talk about atomic swaps, why do we call it atomic? When we are using blockchain systems, because if the two transactions are done in a single block, the block is either confirmed or it's not confirmed. So in that sense, the two transactions are always atomic because they will either both be confirmed they are successful together or both of them are dropped and they are not processed in the first place. So that's what we mean generally when we think about atomic swaps.

But very often today, the assets and payments are not on the same platform. They are on two separate platforms. When that happens, we often have to do much more complicated means of coordinating between the two. So we do have a live project today, what we call payment synchronization, synchronized settlement. So essentially what we do is provide the connectivity features so that when you have another digital asset platform on the other side, we are still able to facilitate DVP settlement. We have done a recent live transaction with Siemens. Again, there are some benefits from that. We are able to settle the transaction in less than a few minutes, probably a lot faster than more conventional traditional ways of doing it. I'll explain a bit more later around how we see the different connectivity model. But what I wanted to point out is that when they are on different platforms, we usually have to have a much more complicated means of allowing that DVP settlement. In this particular case, we use something very similar to a two-phase comic. Essentially, what we have is we have a lock phase and then we have a release phase. So on the cash side, you lock the cash. On the SSR, you lock the assets. Once we confirm that it's locked on both sides, then we release them subsequently.

Essentially, if you are looking at single platform atomic settlement is pretty much just you kind of do it in a single action but when you do it across platforms, you end up with at least four different actions that needs to be done. The other part which I wanted to just lay a bit of a background before we go into. So we find the IMF ASAP model to be very useful when we are looking at all of these different interoperability, connectivity mechanisms. Generally, what it means, how they are looking at it is that you have four layers. At the top you have SS. That's very much like the blockchain wallets or gateways essentially some ability to interact with the underlying services and the assets. And then you have the service layer itself and then the asset layer. So essentially the asset layer will be the different assets like your cash or different digital assets, services essentially some of the additional services, including settlement services for those assets itself. And of course, at the bottom is the platform which you deploy your assets and their services on.

Now, if you look at it from a more technical perspective, we can simplify it down to at the top you have the gateway that connects to your smart contracts, then you have the smart contracts itself which provides like both the assets and the services and then they are all deployed on the blockchain platform. So using that model just now, then very clearly we'll see that there's really three broad means or different approaches to connectivity you could connect them on the gateway level. You could connect them on the blockchain platform level, or you can connect them on the smart contracts level.

Now, when we think about what we are doing, so in the first example that I mentioned, what we are doing today with, you know, like Swire Dakar Bank with DLR, broad reach. So what we are generally looking at is really connectivity on the application layer. Connectivity on the application layer is not specific to blockchain. That's why we are all very familiar with today. You have connectivity on the APIs, two applications having the ability to talk to each other.

Now, API connectivity is something that we are all very familiar with. It's very easy to achieve. Which is why we are generally doing that today.

The other perspective is that, especially in a lot of the enterprise use cases today, you have everyone setting up their own blockchain network. So very often you have one application, one smart contract on a single blockchain network. And because of that application level connectivity is the simplest. You don't actually look beyond that. But what we are starting to see today, the general trend is that we are starting to have enterprise usage of blockchain in a truly shared ledger manner. What I mean by shared leisure is that you can have multiple smart contracts, multiple applications servicing multiple use cases on one blockchain itself.

So the easiest way to think about it, and this is really where we see the value of blockchain. The easiest way to think about it is that if you have two different blockchain platforms on each of the platform itself, you have 100 sets of smart contracts. So if the 100 smart contracts wants to talk among each other, we want to have some ability to enable that. The second part is if 100 smart contracts were to try to talk to 100 different smart contracts on another blockchain itself. Now, if you are looking at an app to app kind of connectivity, essentially that means like 4,950 connections to connect the two the hybrid apps with the other 100 apps.

That is hugely inefficient. So I think what we are generally looking at now is connectivity on a blockchain to blockchain level. Because essentially, if you have some kind of a bridge connectivity between the two, then all your apps have the ability to talk to apps on the other side through the blockchain bridge is a single connection rather than multiple connection. That is an area that we think deserves a lot more effort into developing today.

So in the public blockchain space, generally, when we think about blockchain to blockchain connectivity we think about like blockchain bridges or relayers. Blockchain bridges traditionally have very bad reputation. There's been security vulnerabilities. Part of it is really because in public blockchain, you have two highly secure public blockchain network. But once you are building the bridge, you kind of have to centralized it down to a smaller number of validators. So there are some of these concerns with that, but we think that especially in enterprise blockchain, there's really some value in putting a lot more effort into developing better means of platform to platform level connectivity.

And then the last one, which I was mentioning, is really composability. The idea that if you have smart contracts on a common platform itself. These smart contracts have the ability to directly interact with each other. So this is really the idea behind some of the work that various parties have been pursuing. I'll share a bit more about GL1 in a short while, really just to give a flavor of what we think the future looks like why we are so interested in composability all that ability for the smart contracts on the common blockchain to really interact with each other.

Maybe I'll just pause here to see if there's any questions, any comments around just this general framework that we are using to look at connectivity for blockchain applications.

I guess no questions. I'll continue on. So I wanted to talk a bit more about GR1 itself. From a GL1 perspective, this is really number of banks together with some central banks like MES coming together to say that we see value in blockchain as a shared ledger platform how can we make the most out of it? So from a bank's perspective, we have ourselves, JP Morgan, we have MUFG, We have Citibank. BNY and also subgen so really some of the largest banks around the world coming together to say, how can we make it better? How can we develop something that we can all use?

The use case for this is actually really simple. So from a bank's perspective, we look at what has been happening in the public blockchain world. If you think in terms of like public Ethereum today. Anyone is able to deploy their smart contracts. The smart contracts can interact with each other. The most simplest use case is probably something like you have stablecoins issued by multiple different parties. We have AMM mechanisms, again, developed by a third party. All of these different pieces are actually done by different parties, but yet at the same time, they are able to interact very seamlessly with each other. So it enables a much more integrated ecosystems on public Ethereum.

Now, from a bank perspective, those are really interesting because there's a lot of efficiency gains that we can get out of it. But at the same time, we are very much restricted from using public blockchain today. Very much from a governance perspective because if we need to say. You know, for banks to use any technology, we have to go through say technology risk management, we have to go through outsourcing arrangements, but you're outsourcing arrangements we have to look at who are the parties that we are using now with public blockchain governance of it continue to be an issue for regulator financial institutions because the simplest question what happens if you have another network split? Like what happens between Ethereum and Ethereum Classic? So what happens then?

Who do you actually talk to? Or for example, if for whatever reason something goes wrong on the network, is there someone you can talk to? Is there someone you can work with to get certain things ratified? Now, the way that public blockchains work today, they are intended to be decentralized. That is useful for certain use cases. But from a regulated financial institutions, that is very difficult for us to use and rely on. So our view is that we see value in a much more open and accessible network, but we cannot rely on public blockchain. So what do we do? We decide to come together as a group to let's develop or enhance the technology and subsequently operate this platform such that we are able to use it and deploy our applications.

Now, if you look at what we have drawn up here from a platform asset services and assets layer. Platform level itself, very straightforward. It's going to be another blockchain network. It could be based on any of the existing technology that we see today. We haven't made any decision on

technology itself. Again, the most important part for us is operating the network. We just need to find a technology that works well and possibly if there are gaps. Enhance it such that we are able to use it.

The next level in terms of assets, we see that there's going to be digital currencies, digital assets that could be issued on it. So from our perspective, JP Morgan perspective, we are very interested from a payments perspective. The ability to record issue our liabilities on the platform itself and then to provide payment services to users of the platform? Yes.

Do you still envision this game separated from home? Do you still envision this being separated from identity though, right? So it would essentially be a public blockchain-like environment of transactional data that you as banks that are on off-ramps, et cetera, would have like your KYC still the bank, right? And so I guess my question is is. As a former Gov enforcement person, do you envision a world in which regulators would have direct access to this so that they could do transactional analysis. Or how do you, because that seems like a really great system for doing smarter, not harder AML, CFT, et cetera. Do you envision giving access in some way to regulators, I guess would be my question.

At least to the transactional activity. So from our perspective, we are very happy to work with regulators and with central banks. And it's really, I think we are in the face of really developing that whole concept itself. We continue to interact a lot with regulators and central banks. But the question of whether there's also another question of whether they want to come on board. So that's also an interesting question because.

Just for background, I mean, I used to be in the center. My previous role was actually at the central bank, so I interact with a lot of central bankers and regulators as well. Kind of the problem is always around the issue of moral hazard. Because if a regulator has direct access to all of the information or when something bad happens, how do they continue to take enforcement action because they had the same information, they didn't detect it in the first place.

I think it's a separate conversation from this specific topic altogether, but that continues to be something really interesting that regulators really have to think about, which is do you actually want access to the information and will you act on the information. A lot of times in some of these conversations we actually hear from the regulators that they rather still have they thought being shared with them in the same way that we have always been doing it. But of course, we do see potential value.

Because the whole idea around smart contracts, you could potentially be deploying certain smart contracts on the platform itself. You could actually maybe implement certain rules directly on the platform. You could potentially be running certain set of analyses, doing automated actions when certain things are detected. Some of those rules implementation might actually be a lot easier.

So for example, in some countries they may have rules relating to earn money that says that circulation for any particular wallet cannot exceed a certain amount every year. Now, if you build it into a smart contract, you could potentially stop transactions from happening once those amount has been reached and you need a higher level of KYC or higher level of controls before the wallet access is opened up again. So we do see a lot of potential for what we are doing over here.

From a regulator and central bank perspective, there's a lot of interesting new things that they can do. But ultimately, it's up to the regulators and central banks like, do they want to take advantage of that? Does it actually cost them other related, not technological, but really just from an enforcement perspective of how they actually want to deal with the data.

Oh, does that answer your question? Okay. Thank you. Yeah.

So yeah, so I was talking about the assets itself. So we are envisioning digital currencies, digital assets so some of the banks. They're actually doing a lot of work on digital assets, for example. So repose is a big use case today. That's something that we are doing at JP Morgan itself. Again, we see that because you have all the different banks, they are able to issue their digital currencies, digital assets on a common platform. It allows for a lot more use cases.

Services is really where we start to see a much more clearer identification of the different use cases. Of course, the most simple part of it, payment settlement, security settlement, DVP settlement, PVP settlement, those are the most basic. But once you start layering up, you start to have much more interesting possibilities.

Because you have different currencies there, you can do FX easily. And it doesn't even matter who is that that issuer or distributor of that particular currency. Someone else could be deploying some form of AMM mechanism to facilitate FX settlement. So we start to see much tighter integration of financial services.

You know, many years back we talked about unbundling, rebundling of financial services. I think what we are doing over here can really bring it to an even higher level. And then there's also things like collateral management you could then have repos you could you know pledge some of your collateral and then use that to gain access to liquidity. You could have margin on the platform itself. You could have trade finance, trade, the possibilities are really endless. And I think the really interesting thing is that this could be done by any financial institution on the platform itself, and relying on services already provided by others.

So we see this whole stacking effect where it's very much similar to what we are seeing on public blockchain. For example, some parties come out with different applications and smart contracts. Others are like, "Oh, this is really useful. I don't need to go into that particular area itself. Let me just make use of what someone is already doing really well and build on the next

layer." So again, why we are so interested in this, we see a lot of value from a use case perspective, from an integration perspective. And that's even before we think about the cost perspective.

Because if you look at what a lot of the banks and enterprise blockchain are doing today, they run a single blockchain network for their particular use case. If all of us in the room are running one blockchain network, that's a lot of wasted resources. The concept is very similar to when we first had cloud services, the idea that compute is often elastic. You need it sometimes; you don't need it at other times. Why don't we all come together, run a common platform, and deploy all our applications on it? We save ourselves the hassle and the cost of each running a separate platform. So from an efficiency perspective, we see a lot of value in what we do.

Maybe I'll just pause here. That's an overview of why we are so interested in this idea of GL1 and what the banks are coming together to do. Any questions?

Yes.

Thank you very much for your presentation. During your presentation, you mentioned that you could... So on the payment side... But JP Morgan is like... Financial groups, you could... that we've just been abundant. And I just wonder, for example, other... He's very active in like book nights as a... that's fund and maybe private fund is also expanding. I just wonder how, at that point, you all are grouped to eat this thing.

Thank you. So again, within JP Morgan, we are a large bank. We have interests in many different areas. While my area specifically focuses a lot on payments, we do have other groups within Onyx that focus on tokenized assets. Today, we already report the transaction value per day exceeds more than a billion US dollars. So these are actually live transactions. We are also looking at various forms of tokenized assets. What we are doing today is still confined within regulated financial services. Even when we talk about digital assets, we focus more on the technology aspect, where the underlying financial asset remains the same, but we use technology to make it more efficient.

You're right that we look at it across a wide variety of use cases. In this perspective of GR1, we all come in with a complementary perspective to help us get started. We are taking a deliberate view to focus on payments, while other financial institutions focus on different aspects, making it easier to get started. Ultimately, the banks will compete with each other. From our perspective, we come together to build the underlying blockchain platform as a utility. We don’t see competition at the platform layer as it is commoditized today. Once the platform is built, all the large US banks will be interested in providing US dollar liquidity. When we talk about tokenized assets, again, all banks will be interested. We will see more competition on the platform, which is the intention here.

Ultimately, what we are seeing here is the value for a much more integrated financial ecosystem.

My question is about the public permissioned aspect and how you plan on keeping the assets within the system. In the permissionless space, you can have wrapped Ethereum on multiple layers, collateral here and there, relying on different actors and developers. Do you see this as a competitive advantage if one bank transports assets from GL1 into the public space to access more liquidity? In some wrapping mechanism, wouldn't that defeat the purpose?

That's an interesting question. From a bank perspective, if a bank can use other networks, they would. From a JP Morgan coin perspective, we're also looking at deposit tokens, the idea of issuing our liability on external networks in the future. We don't see the network itself as a competitive element; it's the services provided. So if there's an external network that needs our assets, we'd be happy to issue or distribute on that network if it fulfills all our governance requirements.

In the long run, there might be a divergence initially with assets and currencies on different networks, which may not be efficient due to split liquidity. But with improved connectivity mechanisms, split liquidity may be less of an issue since you can on/off ramp easily. Eventually, there's likely to be convergence, much like in traditional markets. At first, everyone starts with separate OTC transactions, but they later build a marketplace for efficiency. We see the same value here, with all large banks and financial institutions potentially coming together on a common platform.

Ultimately, economics will determine this. We'll likely see divergence before convergence on common platforms.

Thank you. I sort of see a future with GL1 USDC and Ethereum-based USDC, where building a connection between the two just creates competition. I wonder if GL1, as a permissioned platform, can compete with a permissionless marketplace.

Ultimately, it's about who can use it. Today, from a bank's perspective, we can't use a permissionless network. The governance is insufficient, which is why we're creating GR1. This may change in five or ten years if central banks and regulators get comfortable and we address governance issues. We might then have a public network usable by all, but until then, we need GR1 to meet our immediate needs.

Thank you.

We have questions from remote participants. Speak up if you have any questions.

Yes, can I ask a question?

Yes.

I have two questions. First, how do you align accessibility to the money and security sides? As you know, global banks have access to both, while in some jurisdictions, broker-dealers access only securities (CSD), depending on the jurisdiction. How do you plan to align these issues?

From our perspective, GR1 provides the underlying platform, but all financial institutions must comply with laws and regulations. We can't solve legal, regulatory, or governance questions through technology alone. But in the future, if technology can better manage risk—through improved data accessibility or smart contracts, for example—regulators may relax requirements or licensing to take advantage of these efficiencies. However, there's no immediate change on the regulatory front.

Thank you for answering. My second question: who is expected to invest in the development of GL1? It seems like finance and market infrastructure, a public good. If financial institutions invest, they might need to invest in both new and traditional infrastructures. How can we avoid dual investments?

That's a tough question. From an individual financial institution's perspective, there are often dual investments. We invest for the future while enabling immediate benefits. It's up to each institution to decide what's most valuable. GR1 is intended as public infrastructure within the financial ecosystem. Some institutions may want to lead, while others may join later as the network makes sense for them. Membership will allow broader participation, aiming for network effects to make it valuable for everyone.

Thank you. Okay, we'll have a presentation on Project Agula by BOJ later. So, I think these questions don't need immediate answers, but…

I think this is a basic question. When you mentioned interoperability, are you talking about interoperability across apps on GL1 and different platforms or just across apps on GL1?

For assets, you'll have assets on a common platform. One might be a native token, and another might be a smart contract-based token. What will be the native token of GL1, and what is the asset structure?

For interoperability, are you talking about interoperability between GL1 and other platforms?

The first and third questions are related. We see interoperability on multiple layers. For GL1, our focus is on smart contract interoperation on the platform. This seems simple but gets complex with privacy between interacting smart contracts. So, we're addressing composability within GR1.

Ultimately, no single platform will dominate, so platform-to-platform connectivity is crucial, though it's not our immediate focus. The industry should address it.

Okay, so your second question?

I think I have the answer already, so maybe I won't need it.

The next question: Are we discussing interoperability between GL1 and projects like Project Agra?

This is my last point. This is just from our perspective on how we view current initiatives as complementary and potentially convergent in the future.

Today, we have Partio, a multi-bank shared ledger with multiple banks as settlement banks for different currencies. Partio is a full-stack solution as there's no GR1. Each participating bank runs nodes, with connectivity for all participants. Partio works today, but with GR1, Partio could migrate assets and services to GR1. This would centralize the underlying platform while allowing banks to provide services on different layers.

For Project Agora, there's a strong focus on asset and service integration, like central bank and commercial bank money interactions. Agora, and similar initiatives, could deploy on GR1 in the future, gaining connectivity and integration.

I hope that answers your questions.

So, that's all from my side. Happy to take any further questions.

Thank you. Now, we'll move on to the next presentation.

Western, can you speak up now?

Yes, can you hear me?

Yes, I do.

Hello, everyone. My name is Shion Kobayashi from Bank of Japan. I'm a head of the FinTech Center at the Bank of Japan. First of all, thank you for giving me this opportunity to give a presentation of Project Agora. Let me explain what the project Agora is.

The Bank for International Settlements together with seven central banks, including the Bank of Japan, plans to join forces with private sector to explore how organizations can enhance the functioning of the monetary system, the project Agora. Agora means marketplace in Greek. Project Agora builds on the unified ledger concept proposed by the BIS. It will investigate how tokenized commercial bank deposits can be seamlessly integrated with tokenized wholesale central bank money on a public-private programmable financial platform, with the goal of enhancing the functioning of cross-border payments.

The aim is to provide new solutions using smart contracts and programmability while maintaining the two-tier structure of the monetary system. By two-tier, I mean using both central bank money and private bank money.

Project Agora is structured as a public-private collaboration facilitated by the BIS and the Institute of International Finance, IIF. The BIS has brought together seven central banks which are depicted in the left. This group is called the Agora Public Sector Group. We will seek to work in partnership with a large group of private financial institutions convened by the IIF. This group, depicted on the right, is called the Agora Private Sector Group.

Members of the private sector group were selected based on a public call for participation, and members from regulated financial institutions, including but not limited to commercial banks, financial market infrastructures, and critical payment service providers of significance to the cross-border payments ecosystem, were chosen. The selection of members for the private sector was already finalized, and the member list was published last month, including 41 institutions. The project is expected to run through 2025.

The Project Agora committee will guide and direct all project activities and will be co-chaired by the BIS and the IIF. Necessary design and development work is organized into three work streams: business requirements, technology, and legal matters. Additionally, a communications work stream will lead and coordinate all public-facing communications. I myself am a member of the business requirements work stream.

The aim of the project is to test the feasibility of a multi-currency ledger for cross-border payments based on the vision laid out in the "South Internet" paper published by the BIS earlier this year. Project Agora seeks to explore ways to improve the existing financial market infrastructures, which rely on multiple messaging systems and external ledgers for the settlement process.

This will be done by working directly with the commercial banks and private players currently operating in the cross-border payment ecosystem. This figure depicts the typical administrative roles in cross-border transactions, where corresponding banks intervene in the process, such as when currency conversion is involved. The cross-border transaction process often takes longer because each bank has to confirm, for example, the account name and other recipient details. If any problems are discovered in the final process, the procedure may have to be redone from the beginning.

Therefore, Project Agora will investigate solutions and proposals that bring together tokenized commercial bank money and tokenized wholesale central bank money on a programmable shared platform.

In the project, we are very interested in tools such as tokenization. Tokenization has the potential to be the next frontier in a long trend toward digitalization of money. Tokens could combine digital representation of commercial or central bank money with programmable instructions to streamline processes and simplify complex tasks. We believe this can unlock types of contracts and transactions that are not viable today.

Project Agora's main objective is to demonstrate how a unified ledger could enhance the efficiency of business and regulatory processes in correspondent banking payment chains, thereby reducing transaction times and costs, enhancing payment transparency, and mitigating risks for banks involved in cross-border payments.

The project aims to build a technical prototype to test wholesale cross-border payments and demonstrate the following: the ability to represent tokenized commercial and wholesale central money on a unified ledger that meets technical and business requirements, atomic cross-border transactions between currency areas involving a single currency pair, the ability to streamline pre-validation efforts to reduce duplication without changing existing standards, and improved efficiencies in complying with AML, CFT, and other functional screening.

Project Agora also seeks to analyze and identify potential legal and regulatory issues and gaps across all seven jurisdictions related to tokenized commercial and wholesale central bank money. Project Agora aims to go beyond proof of concept and deliver a prototype to test a range of potential current and future use cases. The lessons learned may set a path toward a new type of financial market infrastructure based on new technology.

As already mentioned, the project is expected to run through 2025, making the next 14 months crucial. Since we are still working on fleshing out the details of the use cases, the technical workstream discussion has not yet begun in earnest. Please understand that as of today, no one has answers to technical questions about interoperability with Project Agora.

Before I end my talk, I would like to add something regarding the IMF ASAP model, which Wiki also commented on. The ASAP model is a conceptual model that allows for a deeper understanding of the various developments underway by categorizing them into four layers: assets, services, applications, and platform. Global Layer 1 is focused on creating a new platform, while Project Agora focuses on the asset and service layers, specifically the coexistence of central bank and tokenized commercial bank money. In this regard, Global Layer 1 focuses on the platform, while Project Agora focuses on asset and service layers.

Thank you for your attention. Are there any questions from the audience?

Maybe you have already talked about it a little bit, but I think I covered some of these things in the sharing earlier. So broadly, the way we think about it is connectivity could be done on

different layers itself. Between applications, it's really API connectivity, which I think is very well established today, so there's not much to think about.

From a smart contract composability perspective, I think some of the questions we need to start thinking deeper about are: how do we actually enable multiple smart contracts to interact with each other while still preserving privacy? That's probably one of the most difficult questions.

To explain a bit further, let's say there are multiple transactions that need linking, but each part of the transaction itself has a separate group of parties. If we think about the usual privacy manager approach, it doesn't work very well because you either have all parties as part of the same privacy group, giving them access to information irrelevant to them, or you don't have a good ability to pass only the related information. From a technical perspective, a lot more needs to be explored there.

From a platform-to-platform level connectivity, things we've been looking at include bridges, relayers, and layer-one, layer-two relationships, which also require a lot more work, especially from a technical perspective. Lastly, there's also wallet interoperability. I understand that some parties are exploring universal wallet infrastructure, which is also an interesting aspect to consider.

My question is about the interoperability between smart contracts. You mentioned a privacy problem in terms of transaction interoperability. Could you dive deeper into this aspect? I'm not really sure what kind of privacy you mean when it comes to interoperability of smart contracts.

In terms of smart contract interoperability, privacy is essential. For example, with asset settlement and payment settlement, the transacting parties are the same on both sides, involving the sender, recipient, or buyer and seller. But on the payment side, you might have a bank, while on the asset side, there's an exchange or regulated market operator, essentially a different party that's not the bank.

The bank doesn't need to know all details of the asset side; it just needs enough information for the linked transaction. Privacy managers like BESU's Tessera don't work well here because they make all transaction details private among parties. But if all six parties are in the privacy group, it allows the transaction yet reveals more information than necessary.

That is a hard technical problem: enabling composability while maintaining privacy. It's a question that we need to solve on the platform level. Another interesting problem is orchestrating more than two smart contracts. With two contracts, you simply have one calling the other. But with three contracts, how do you pass information between, say, A and C without going through B?

The easiest approach might be an orchestrator smart contract, but this brings up more questions: how do you map the technical aspects to the business? Typically, each smart contract represents a licensed activity performed by an entity, so you know who's responsible. With an orchestrated smart contract, there's often no direct linkage to a legal entity. Broadly, these are some of the challenges we're looking at in GL1, focusing on composability but recognizing that these problems affect the broader ecosystem.

I have a question about technical standards for tokenization. When tokenizing things without an agreed value, like stablecoins or fiat between banks, what are some challenges? Especially with less liquid assets, what role do pricing vendors play?

If you're considering securities, when you have an asset token, it essentially represents the units of that asset. Cash works similarly since it's priced in its currency. From a token standard perspective, I don't think pricing is an immediate need—it's more of a business use case requirement.

For example, using digital assets for collateral management requires accurate prices for under-collateralization. This is more about business use cases than token standards. The real challenge in token standards is separating what goes on-chain from what stays off-chain while still allowing tagging between the two.

In payments today, there are well-established messages like MT or ISO 20022 standards, which capture a lot of information not intended for the blockchain. Settlement involves debiting one account and crediting another with certain information transmitted off-chain for payments control, sanction screening, and AML/CFT processes. The difficulty is separating these on-chain and off-chain parts, managing their flow, and changing the interaction flow for broader information capture.

Early ideas around this involve whitelisting and verifiable credentials. But these issues need ecosystem-level solutions and standard-setting bodies. The discussion can involve either specific technical solutions or general governance on bringing parties together.

Could you talk about cybersecurity aspects in the project, especially cryptographic considerations?

We don't have specific answers here. Most of this is handled on the underlying technology level. We have cryptographers who review and address vulnerabilities, but these aspects are often abstracted from developers. We do have cybersecurity experts who review the technology.

Okay, this particular new technology that we are evaluating—how well does it work? Again, I don't have a perfect answer, just wanted to share my perspective. I'm sure there are other experts here who might want to chime in.

Great. So, in terms of technical standards and interoperability standards, Thomas, are you here now? Could you talk a bit about what you do in IETF, particularly on technical standards for the interoperability of DLT technologies? I think it would be helpful if you could discuss this to start the conversation on technical standards and interoperability.

Sure, happy to. Can you hear me okay? Yes, we can hear you. Great. For those unfamiliar, the IETF is a community-driven standards organization responsible for developing 99% of the protocols we use daily—SSL, TLS, HTTP, and identity management protocols for logging in with Gmail or Facebook. It's been around for about 35 years.

One working group there, created a couple of years back, is the Secure Asset Transfer Protocol Working Group. Its goal is to build standard components and APIs that projects can utilize to achieve a degree of interoperability for common functions. There's a lot of synergy here with what was previously discussed about models, bridges, and composability. The IETF doesn't address complete solutions but provides piecemeal components for broad interoperability.

The Secure Asset Transfer Protocol Working Group currently has three main work items. One is a common interoperability architecture that provides a background, high-level overview. The second is the SAP PCOR, a unidirectional asset transfer protocol based on a burn-mint model, asset-neutral and agnostic to the underlying technology. For example, you can use it for a DVP (delivery versus payment) by running two instances, one for asset transfer and the other for payment, in a currency like a CBDC or stablecoin.

IETF meets three times yearly with around 2,000 engineers attending across various working groups, from routing to video codec standards. Our next meeting is in Dublin, open to anyone without membership requirements. Meetings rotate between Asia, North America, and Europe. Joining the mailing list is also an option to participate passively.

The IETF Secure Asset Transfer Protocol Working Group aims to standardize components for interoperability across diverse assets. We're developing a schema to define assets universally so both sides of a transaction can recognize the same asset across different blockchains. We're also considering asset-swapping standards, which is a high priority next year. The aim is to reduce custom implementation per network—one integration to serve all.

Active participation in the IETF provides valuable insight and allows us to create standards that benefit the entire ecosystem. We invite all interested to join our discussions, mailing lists, or meetings.

To maintain three years of collaboration, the key is alignment among organizations. Large entities, governments, and financial institutions want standardization to guarantee interoperability. Each IETF working group strives for a "build once, deploy everywhere" approach, which benefits members by reducing custom implementations and simplifying asset

transfers. A spec and at least one open-source implementation are required for completion, ensuring open standards and deployment.

The purpose isn’t to make money on plumbing but to make settlement faster and seamless. Please join our mailing list or reach out directly if you want to discuss further.

Okay, thank you, Thomas. I'll stop because we don't have any more time. let's finish up the session and thank you everyone for joining this session